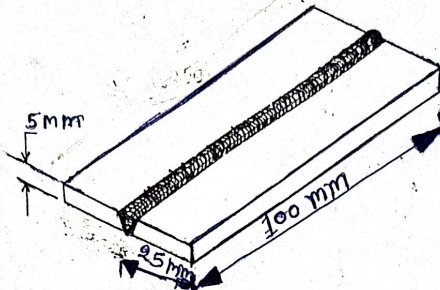


WELDING SHOP

PRACTICE-I

Time : 3 Hrs.

OBJECT:- To make a simple butt joint by electric arc welding process as per given drawing.



MATERIAL REQUIRED:- M.S. Flat 100mm x 25mm x 5mm - 2Nos.
M.S. Coated electrode 10 SWG.

TOOLS AND EQUIPMENTS USED:-

1) **Holding Tools:**

- a) Flat tong

2) **Cutting Tools:**

- a) Hacksaw

3) **Slag and Spatter Removing Tools:**

- a) Chipping Hammer
- b) Steel wire brush

4) **Welding Machine:**

- a) *Reck for m/c*
A.C. Arc welding step down transformer set
- b) D.C. Arc welding motor generator set

5) **Electrode Holder:**

6) **Miscellaneous Equipments:**

- a) Working Table
- b) Earthing Clamp

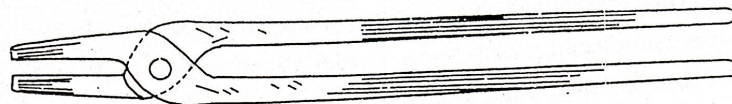
7) **Safety Equipments:**

- a) Welding screen
- b) Leather Apron
- c) Leather Gloves

DESCRIPTION OF TOOLS AND EQUIPMENTS:-

1. Flat Tong :

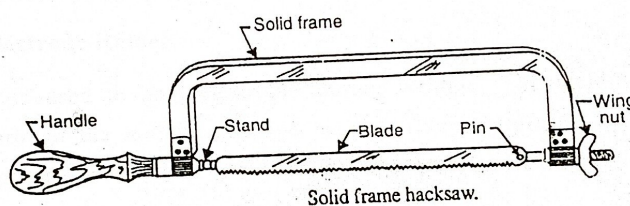
It is made of mild steel and used for holding the hot job.



flat tong

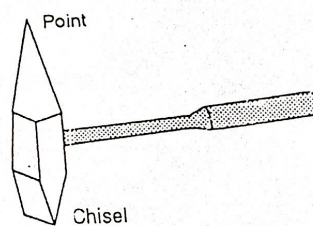
2. Hacksaw :

The Hacksaw is the chief tool used for cutting flats, rods, and pipe in to desired lengths. It consists of a metal frame which may be solid or adjustable. The blades fits over two pegs which protect from the pins sliding in the ends of the frame. The blades are made of carbon or high speed steel and may be finished with the cutting edge only hardened or they may be hard all over.



3. a) Chipping Hammer :

These are hammers having pointed or sharp striking face and used for removing slag and spatter from the work pieces after welding.



b) Steel Wire Brush :

It is used for removing slag and unwanted materials from metal surface in fine way than chipping hammer.



Wire Brush

4. a) **A.C. Arc Welding Step Down Transformer Set :**

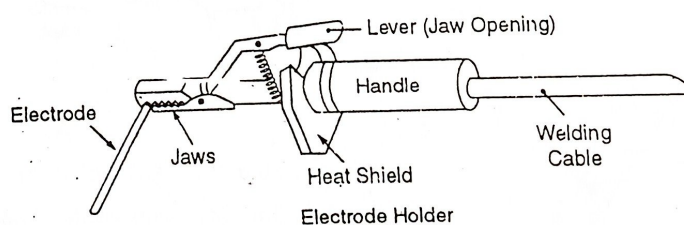
The A.C. Arc welding step down transformer built in with the welding plant and help in reducing the voltage from 440 volt to about 80 volt. The transformer of the welding plant may be of single phase type or multiphase type depending on the main power supply available. Generally for small fabrication jobs or repair jobs, a single phase transformer welding set is preferred but for higher working load a three phase transformer welding plant is used. It has no moving parts and hence possesses low maintenance cost and low installation cost.

b) **D.C. Arc Welding Motor Generator Set :**

The D.C. Welding machine are either (a) Motor generator type or (b) AC-DC transformer rectifier type. One of the big advantage of a DC welding set is that either straight or reverse polarity can be used. Polarity indicates the direction of current flow. Polarity can be changed by a switch on the welding machine or by changing the cable connections. Most of the heat is liberated in the positive side of the arc.

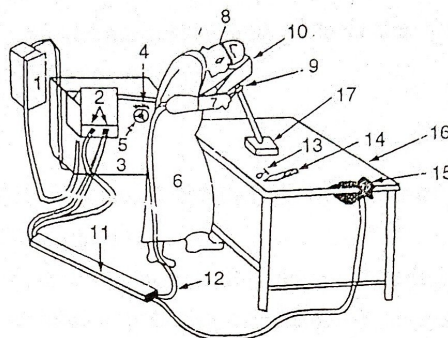
5. **Electrode Holder :**

It is used to hold electrodes during welding. The welding cable passes through the hollow insulated handle. The electrode is held between two spring loaded jaws. Electrode holders are also rated in terms of current carrying capacity.



6. a) Working Table :

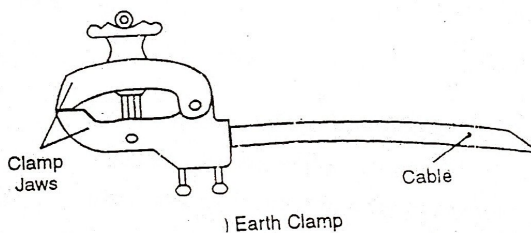
It is made of mild steel and the job is placed on the working table during welding. The earthing clamp is clamped in working table.



Arc welding set-up (1) Switch box, (2) Secondary terminals, (3) Welding machine, (4) Current reading scale, (5) Current regulating hand-wheel, (6) Leather apron, (7) Leather hand gloves, (8) Protective glasses strap, (9) Electrode holder, (10) Hand shield, (11) Channel for cable protection, (12) Welding cable, (13) Chipping hammer, (14) Wire brush, (15) Earth clamp, (16) Welding table (metallic), (17) Job.

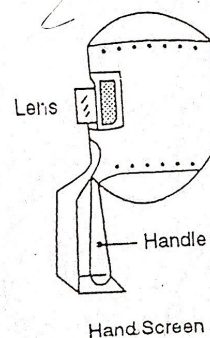
b) Earthing Clamp :

It connects the earthing cable to work-piece either directly or through the welding table on which work piece is placed.



7 a) Welding Screen :

It is used to protect eye and face from spark and ultraviolet and infrared rays emerging from the arc during welding. It is made up of fibre sheet and black glass are fitted to look at weld.



b) Leather Apron :

Aprons are made of chrome leather they are used for protection against spatter and spark etc.

c) Leather Gloves :

These are made of chrome leather and protects the hand from spark and spattering of metal during welding.

PROCEDURE :-

1. Cut the two pieces of required length from M.S. Flat of required width and thickness by means of hacksaw
2. Connect the earthing plug with welding table consisting iron top and three side cabin like structures to escape dispersion of committed rays and light etc. out of work place.
3. Set current (90-120 Amp.) and voltage (20 Volt) according to the thickness of job.
4. Place the mild steel job on working table in proper position after making it clean with the help of wire brush.
5. Hold the electrode in electrode holder.
6. Start the welding from upper side to lower side at a uniform speed.
7. Remove the slag and spatters etc. with the help of chipping hammer and steel wire brush.

PRECAUTIONS:-

1. Check all the connections before starting the welding.
2. Use suitable current and voltage values for welding.
3. Always adopt right tool and right procedure for every operation in welding shop.
4. Always use leather apron, gloves and welding screen during welding operation. Welding should never be seen with naked eyes during welding.
5. Avoid stacking electrode at job in arc welding
6. Avoid chatting with others while doing welding.
7. Always select correct electrode size for welding.
8. Always use tongs to hold the hot work piece.
9. Switch off the main switch when the work of the day is over. This is done in order to avoid any short circuit, sparking etc.